

Mohan Birajdar

Full-Stack Developer

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· Pune, Maharashtra

PROFESSIONAL SUMMARY

I'm a Software Engineer who enjoys building scalable backend systems, distributed applications, and production-grade AI solutions that solve real-world problems. I love turning complex ideas into reliable software—from designing responsive React applications to developing high-performance backend services with Node.js, Python, and Java. Over the past year, I've worked on enterprise healthcare platforms, multi-agent AI systems, Retrieval-Augmented Generation (RAG) pipelines, and secure cloud-ready applications. What excites me most isn't just building new features—it's understanding why systems fail in production, how they scale under real-world load, and how thoughtful engineering can make software more reliable, maintainable, and trustworthy. CGPA **9.3** · AI/ML Minor · MIT Aurangabad.

TECHNICAL SKILLS

AI & Machine Learning	Agentic AI, Multi-Agent Systems, Retrieval-Augmented Generation (RAG), LLMs, LangChain, Vector Databases, Embeddings, Prompt Engineering, Model Evaluation, CTGAN, TVAE.
Languages	TypeScript, JavaScript (ES6+), Python, Java
Frontend	React.js, Next.js, HTML5, CSS3, Tailwind CSS, Bootstrap, Responsive Web Design, State Management (Zustand)
Backend / APIs	Node.js, Express.js, Laravel, PHP, FastAPI, RESTful APIs, Microservices, API Development, MVC Architecture
Databases	PostgreSQL, MongoDB, MySQL, Supabase, Prisma ORM
Security	AES-256-GCM encryption, RBAC, JWT Auth, Audit Logging
Cloud / DevOps	Docker, Git, GitHub, GitHub Actions, CI/CD, Cloud Deployment, Vercel
Software Engineering	Data Structures & Algorithms, Object-Oriented Programming (OOP), SOLID Principles, System Design, Distributed Systems, Scalable Architecture, MVC, REST Architecture, SDLC, Agile (Scrum), Performance Optimization, Debugging, Problem Solving

PROFESSIONAL EXPERIENCE

SoftGrid Info Pvt. Ltd.

Software Engineer Trainee · Pune

July 2026 – Present

- Contributing to full-stack web application development using the **MERN stack** while developing backend features for the **BrightTech** educational platform with **Laravel**.
- Building RESTful APIs, debugging and optimizing application modules, and supporting production deployments.
- Collaborating with cross-functional teams using Git and Agile practices to deliver scalable and maintainable software solutions.

Symetrique Inc

Feb 2025 – June 2026(over 11 months)

Software Developer (Full-Stack) & Data Analyst | AI Engineer · Chicago, USA (Remote)

Symetrique is a healthcare intelligence company that turns fragmented real-world data — claims, EHR/EMR, labs, registries — into governed, decision-grade evidence for payers, biopharma, and HTA bodies. I've grown through two roles here, starting as a data analyst and evolving into a full AI engineering function.

Developing production-grade healthcare software by building scalable full-stack applications, backend services, secure APIs, AI-powered workflows, and data platforms. Contributing across the complete software development lifecycle from system design and implementation to deployment and maintenance.

- Developed ClinKit, an enterprise clinical trial management platform with 8 integrated modules for inventory management, demand forecasting, expiry monitoring, and workflow automation across 10 hospitals in 6 countries.

- Built VITARA, a secure healthcare platform using React (Next.js), TypeScript, Node.js, Supabase, and Prisma with responsive UI, REST API integration, secure authentication, RBAC, and scalable application architecture.
- Designed Python-based data engineering pipelines using CTGAN and TVAE to generate over 1,500 synthetic healthcare records with complete schema validation for secure product development and testing.
- Developed RESTful backend services and intelligent automation workflows supporting healthcare applications, improving operational efficiency while maintaining security, scalability, and reliability.
- Engineered a subscription-based SaaS platform using Laravel, MySQL, REST APIs, PayPal integration, and role-based access control, automating user provisioning and payment-gated feature access.
- Built interactive dashboards and automated analytics pipelines using Python, Pandas, and NumPy, reducing manual reporting effort by 25% and improving decision-making through real-time insights.

ROJECTS

NextStepDoctor — AI-Powered Medical Research Assistant

Apr 2026

next-step-doctor.vercel.app

Production-ready full-stack healthcare platform delivering research-backed medical insights through a responsive React/Next.js application and scalable backend services.

Stack: React.js, Next.js 15, TypeScript, Node.js, Express.js, FastAPI (Python), MongoDB, REST APIs, RAG, LLaMA 3.1, AES-256-GCM, Zustand, Whisper

- **Design thesis:** Designed and developed a responsive full-stack application using React, Next.js, TypeScript, and Node.js with reusable components, REST API integration, secure authentication, and scalable backend architecture.
- **8-Stage RAG Architecture:** Designed scalable backend services using Node.js and FastAPI to orchestrate an 8-stage RAG pipeline that retrieves, ranks, and validates medical evidence from multiple trusted healthcare sources.
- **Agentic Copilot with HITL:** Monitors full conversation context and proactively proposes structured health-profile updates (medications, diagnoses, allergy flags) through a human-in-the-loop confirmation step — treating user health data as living state.
- **Privacy & Accessibility:** Strengthened application security through AES-256-GCM encryption, RBAC, JWT authentication, audit logging, and HIPAA-aware architecture while optimizing maintainability and production reliability.

VeriFlow AI — Multi-Agent SAP Compliance Engine

June 2026

Production-grade enterprise platform that automates SAP compliance auditing using scalable backend services, distributed AI workflows, and Retrieval-Augmented Generation (RAG), reducing manual document review from hours to minutes.

Stack: Node.js, Express.js, TypeScript, MongoDB, REST APIs, Vector Database, RAG, Groq API (LLaMA 3.3 70B)

- **Architecture:** **DocumentAgent** (PDF parsing, structured extraction) → **RagAgent** (vector similarity retrieval, chunking tuned to regulatory docs) → **ComplianceAgent** (chain-of-thought PASS/FAIL/WARNING) → **ReportAgent** (scored compliance report). Each agent has a defined I/O contract — auditable and extensible.
- **Scalability:** Persistent rule-library storage in MongoDB eliminates re-upload overhead between audit runs. RagAgent chunking tuned to regulatory document structure improves retrieval precision over naive full-document embedding.
- **Reliability:** Enhanced production reliability through automatic model failover, retry strategies with exponential backoff, structured error propagation, and centralized logging, improving observability, fault tolerance, and overall system resilience.

EDUCATION

Maharashtra Institute of Technology, Chh. Sambhajinagar

June 2022 – May 2026

B.Tech — Electronics & Computer Engineering · CGPA: 9.3 / 10.0

Minor Specialization: Artificial Intelligence & Machine Learning (Grade: A)

ACHIEVEMENT

- CGPA 9.3/10 with a Minor in Artificial Intelligence & Machine Learning.
- Built production-grade AI and Full-Stack applications across healthcare, compliance, and education domains.
- Elite – NPTEL Cloud Computing (IIT Kharagpur) | NPTEL Programming in Java Certified.
- Delivered **2 freelance web applications** (*Ras Dandiya* and *Om Sai Bhojnalay*) for clients on a fixed-price basis.